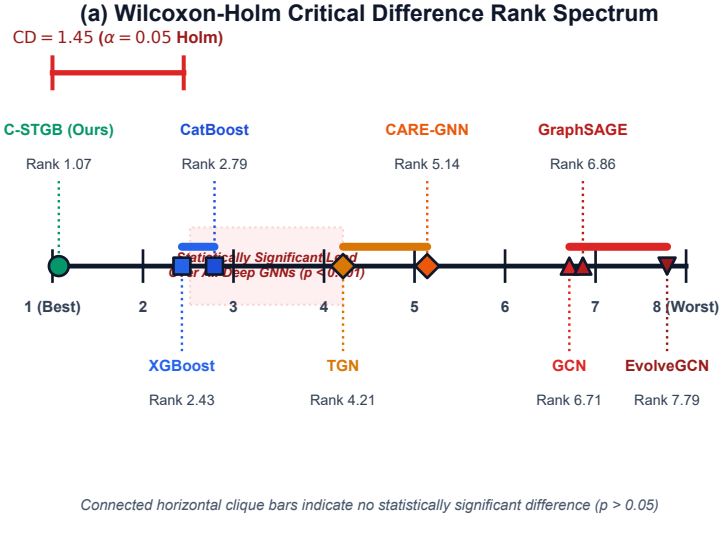
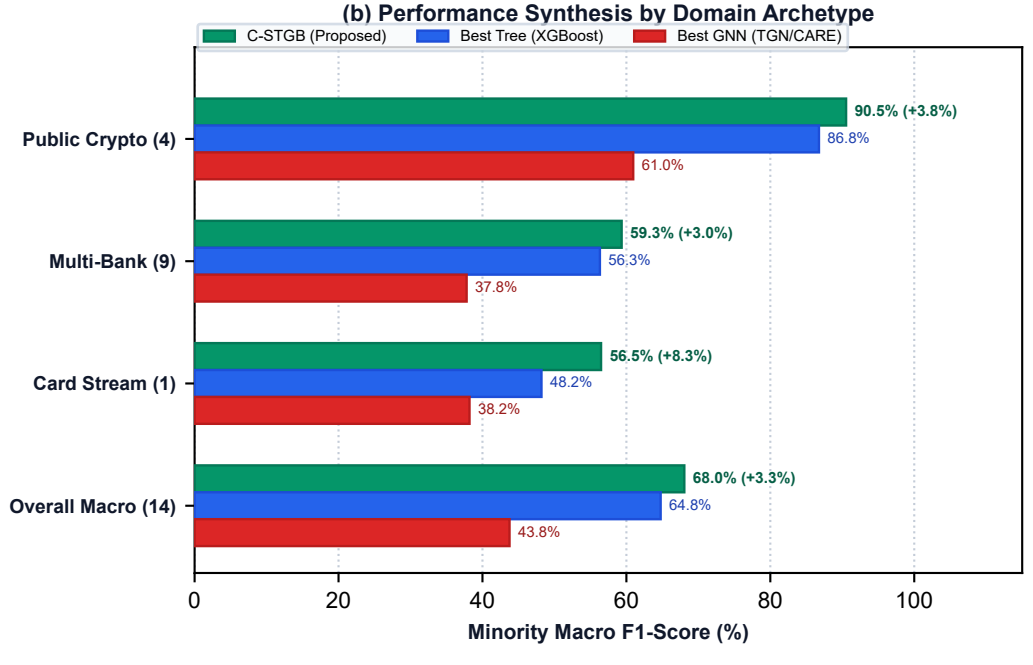

(a) Wilcoxon-Holm Critical Difference Rank Spectrum
CD = 1.45 (α = 0.05 Holm)
C-STGB (Ours)
Rank 1.07
CatBoost
Rank 2.79
CARE-GNN
Rank 5.14
GraphSAGE
Rank 6.86
Statistically Significant Lead
over All Deep GNNs (p < 0.01)
1 (Best)
2
3
4
5
6
7
8 (Worst)
XGBoost
Rank 2.43
TGN
Rank 4.21
GCN
Rank 6.71
EvolveGCN
Rank 7.79
Connected horizontal clique bars indicate no statistically significant difference (p > 0.05)


(b) Performance Synthesis by Domain Archetype
C-STGB (Proposed)
Best Tree (XGBoost)
Best GNN (TGN/CARE)
Public Crypto (4)
90.5% (+3.8%)
86.8%
61.0%
Multi-Bank (9)
59.3% (+3.0%)
56.3%
37.8%
Card Stream (1)
56.5% (+8.3%)
48.2%
38.2%
Overall Macro (14)
68.0% (+3.3%)
64.8%
43.8%
0
20
40
60
80
100
Minority Macro F1-Score (%)